GHXTox: Peptide toxicity prediction by fusing a task-adaptive protein language model with pLDDT-aware multi-scale structure

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Abstract

As biomedical research advances from the cellular to the molecular level, peptides have become increasingly important as a versatile class of bioactive molecules and are now widely used in the targeted treatment of diseases such as Parkinson's disease and cancer. However, toxicity assessment remains a critical bottleneck in peptide-based drug development. Existing methods have begun to explore multi-dimensional (multimodal) feature fusion and toxicity prediction based on the three-dimensional (3D) conformations of peptides, yet most of them still underutilize 3D structural information, and systematic multi-dimensional fusion remains rare. Here, we propose GHXTox, a hierarchical dual-level fusion framework. In the first level, a sequence-enhanced 3D structural expert integrates sequence semantics with spatial structural features through pLDDT-aware cross-attention. In the second level, this structural expert is fused with a task-adaptive one-dimensional (1D) sequence expert at the probability level, using weights frozen on training out-of-fold (OOF) predictions. The 1D branch analyzes sequences with a task-adaptive protein language model (ProtT5 fine-tuned with low-rank adaptation, LoRA) and a bidirectional Mamba state-space model. The 3D branch extracts multi-scale structural information from predicted conformations, from local residue functional groups to the whole peptide, and exploits the predicted local distance difference test (pLDDT) confidence through quality-aware gating. GHXTox achieves competitive performance on independent test sets and provides more detailed biological interpretations for peptide toxicity prediction.

Keywords: peptide toxicity prediction; multimodal deep learning; protein language model; bidirectional Mamba; pLDDT-aware conformer ensemble; interpretability

1 INTRODUCTION

As biomedical research advances from the cellular to the molecular level, peptides have become increasingly important as a versatile class of bioactive molecules and are now widely used in the targeted treatment of diseases such as Parkinson's disease and cancer [1]. Peptide toxicity is a critical safety consideration throughout the drug development pipeline, and the accurate and efficient identification of toxic peptides remains a major bottleneck in peptide-based drug discovery [2]. Because conventional experimental toxicity assays are costly, time-consuming, and difficult to scale, reliable computational predictors are highly desirable.

Computational peptide toxicity prediction has evolved from traditional machine learning methods based on handcrafted physicochemical features toward deep learning approaches with stronger representation-learning capability. Current mainstream methods can be roughly grouped into two categories. The first category operates on sequence-level (one-dimensional, 1D) and/or atomic-level (two-dimensional, 2D) representations. For example, ToxMSRC combines continuous bag-of-words (CBOW) embeddings, multi-scale convolutional neural networks, and a bidirectional long short-term memory network with residual connections, achieving strong performance on the benchmark dataset [2]. ToxPLTC integrates pooled ProtT5 embeddings, borderline-SMOTE augmentation, and a text convolutional neural network [3]. ToxiPep fuses context-aware sequence representations with atomic-level graphs through a cross-attention mechanism [4]. HyPepTox-Fuse further combines embeddings from multiple protein language models

with conventional descriptors [5]. Although these models achieve strong performance within their respective modalities, they largely neglect the three-dimensional (3D) conformations of peptides, which are closely related to toxicity; cross-dimensional information interaction and mutual validation remain limited, and their interpretability is mostly confined to the sequence level.

The second category attempts to exploit 3D structural information. PeptiTox uses ESM2 sequence embeddings together with ESMFold-predicted 3D structures and geometric graph neural networks [6]. StrucToxNet feeds ProtT5 sequence embeddings and ESMFold-predicted structures into an equivariant graph neural network (EGNN) and systematically analyzes structural quality [7]. These studies demonstrate the value of 3D conformations for toxicity prediction. However, such methods are constrained by the limited accuracy of current structure prediction models on short peptides and typically rely on a single predicted conformation; structural confidence, i.e., the predicted local distance difference test (pLDDT) score, is either discarded through hard filtering or ignored, and multi-conformer ensembles are not considered. Moreover, the fusion between the sequence and structure branches remains relatively shallow, and no systematic multi-dimensional fusion that combines a task-adaptive sequence model with confidence-aware multi-scale structure modeling has been established.

To address these limitations, we propose GHXTox, an interpretable hierarchical dual-level fusion framework for peptide toxicity prediction. In the first level, a sequence-enhanced 3D structural expert integrates sequence semantics with spatial structural features through pLDDT-aware cross-attention. In the second level, this structural expert is fused with a task-adaptive 1D sequence expert at the probability level, with weights frozen on training out-of-fold (OOF) predictions. In the 1D branch, a task-adaptive protein language model (ProtT5 fine-tuned with low-rank adaptation, LoRA) is combined with a bidirectional Mamba state-space model for sequence analysis [8–10]. In the 3D branch, because current structure prediction models generally output a single predicted conformation with limited accuracy on short peptides, we adopt a pragmatic structure augmentation scheme in which different structure prediction and conformational sampling models are applied across different sequence-length ranges [11–14]. On this basis, we design a multi-scale structural information extraction strategy that progressively captures structural cues from local residue functional groups (chemical sites) to the whole peptide, and the pLDDT confidence output by structure prediction models is exploited through continuous down-weighting and dynamic gating of structural features. The 1D ensemble probability is first calibrated using training out-of-fold predictions, after which the calibrated 1D probability and the structural-expert probability are combined using OOF-frozen fusion weights. Ablation experiments demonstrate that the two dimensions complement each other effectively and that the fusion substantially improves overall performance, while interpretability analyses provide more detailed biological explanations for the predictions.

In summary, the main contributions of this work are threefold: (1) we propose GHXTox, an interpretable hierarchical dual-level fusion framework that integrates a task-adaptive 1D protein language model with pLDDT-aware multi-scale 3D structure modeling; (2) the 3D branch designs a multi-scale structural information extraction strategy spanning from local residue functional groups to the whole peptide, and incorporates pLDDT confidence into structural feature processing through continuous down-weighting and dynamic gating; and (3) extensive experiments on benchmark and independent test sets show that GHXTox achieves competitive performance, and ablation, robustness, and interpretability analyses confirm the effectiveness and complementarity of the two dimensions.

2 MATERIALS AND METHODS

2.1 Datasets

We evaluate GHXTox on the benchmark datasets constructed by CAPTP [17] and subsequently adopted by ToxMSRC [2]. Following the data construction protocol of ToxMSRC [2], toxic peptide sequences were merged from three databases, namely CSM-Toxin [18], ToxinPred2 [19], and ATSE [20], and sequences longer than 50 residues were removed. Non-toxic sequences were retrieved from UniProt [21] using the query "NOT KW-0800 AND NOT KW-0020 AND reviewed:true", and sequences containing non-standard amino acids or exceeding 50 residues were excluded. After combining the two groups and removing samples with conflicting labels, CD-HIT [22] was applied with a sequence-identity threshold of 0.9 to eliminate redundant sequences, yielding 2,138 toxic and 5,375 non-toxic peptides. Following the same partition as ToxMSRC [2], we used 85% of the data as the training set and the remaining 15% as the independent Test1 set, resulting in 1,818 toxic and 4,569 non-toxic training peptides and 320 toxic and 806 non-toxic Test1 peptides. An additional independent Test2 set, containing 46 toxic and 536 non-toxic peptides, was used to further evaluate generalization ability. Table 1 summarizes the statistics of the datasets used in this study.

Table 1: Statistics of the datasets used in this study.

Dataset	Toxic	Non-toxic	Total	Use
Training (benchmark)	1,818	4,569	6,387	Model training
Test1 (benchmark)	320	806	1,126	Independent test
Test2	46	536	582	Independent test

2.2 Overview of the Model

Figure 1 shows an overview of GHXTox, a hierarchical dual-level fusion framework for peptide toxicity prediction. The framework consists of a task-adaptive one-dimensional (1D) sequence expert, a sequence-enhanced three-dimensional (3D) structural expert, and a probability fusion module driven by training out-of-fold (OOF) predictions. Within the structural expert, GHXTox first performs feature-level interaction between sequence semantics and spatial structural information through pLDDT-aware cross-attention. At the top level of the model, the prediction probabilities produced by the 1D sequence expert and the 3D structural expert are then fused at the decision level. GHXTox therefore combines intra-expert feature-level fusion with inter-expert decision-level fusion, preserving the independent discriminative capability of each branch while exploiting complementary information from amino acid sequences, spatial geometry, local chemical environments, and conformational variability.

Figure 1: Overall architecture of GHXTox. GHXTox adopts a hierarchical dual-level fusion design. The task-adaptive 1D sequence expert extracts toxicity-related sequence features with a LoRA-fine-tuned ProtT5 encoder and a bidirectional Mamba network, followed by three-seed ensembling and out-of-fold (OOF) Platt calibration to produce p_{1D}. The sequence-enhanced 3D structural expert extracts multi-scale structural information through an ESM2-enhanced sequence encoder, a pLDDT-aware equivariant graph neural network, full-atom chemical-site modeling, and an availability-masked conformer residual branch. Within this expert, sequence and spatial residue representations are integrated through pLDDT-aware cross-attention to produce p_{SE3D}. The two expert probabilities are fused with frozen weights of 0.72 and 0.28, and the final toxicity decision is obtained with a threshold of 0.4702.

2.2.1 Task-Adaptive 1D Sequence Expert

The task-adaptive 1D sequence expert takes the amino acid sequence of a peptide as input. After tokenization, the sequence is processed by the ProtT5 protein language model [CITE: ProtT5, Elnaggar et al. 2021] to obtain contextualized, residue-level representations. To adapt this general-purpose protein language model to peptide toxicity prediction, we employ low-rank adaptation (LoRA) [CITE: LoRA, Hu et al. 2022] for parameter-efficient fine-tuning, learning task-specific updates while the majority of the pretrained backbone parameters remain frozen. The resulting residue-level representations are then fed into a bidirectional Mamba state-space network [CITE: Mamba, Gu and Dao 2023], which models residue dependencies in both the N-to-C and C-to-N directions. After residual connections, layer normalization, and masked pooling, a classification head produces the sequence-level toxicity probability. To reduce the variance arising from random initialization, we train three models on the full training set with different random seeds and average their output probabilities. The ensemble probability is then calibrated with Platt scaling [CITE: Platt 1999] fitted exclusively on training out-of-fold (OOF) predictions, yielding the final 1D expert probability $p_{1\mathrm{D}}$.

2.2.2 Sequence-Enhanced 3D Structural Expert

The sequence-enhanced 3D structural expert comprises four subbranches: an internal sequence-semantic subbranch, a spatial-geometric subbranch, a full-atom chemical-site subbranch, and an availability-masked conformer-ensemble residual subbranch. Collectively, these components cover five complementary representation scales: residue-level semantics, atom-level chemical sites, spatial residue neighborhoods, peptide-level global properties, and conformer-ensemble statistics. In the sequence-semantic subbranch, amino acid residue types, functional-group properties, explicit physicochemical descriptors, and ESM2 residue embeddings [CITE: ESM2] are projected into a shared feature space and then processed by a two-layer Transformer encoder [CITE: Transformer] to generate contextualized sequence representations. In parallel, the spatial-geometric subbranch constructs a hybrid residue graph that combines sequential adjacency edges with spatial-neighborhood edges derived from the predicted peptide structure. Residue physicochemical properties, local geometric descriptors, ESM2 embeddings, three-dimensional coordinates, and residue-level pLDDT scores are then fed into three pLDDT-aware equivariant graph neural network (EGNN) [CITE: EGNN] layers to update the residue states and spatial representations.

To characterize local chemical environments that cannot be fully captured by a C_α-based residue graph alone, we extract chemical sites corresponding to charged, aromatic, and hydrophobic groups as well as hydrogen-bond donors and acceptors from the full-atom peptide structure. Message passing [CITE: MPNN, Gilmer et al. 2017] is then performed over these sites using pairwise distances, directional relationships, interaction types, and structural confidence. The resulting chemical-site features are mapped back into the corresponding residue-level representations through a zero-initialized residual projection. This design allows the model to progressively learn toxicity-related local spatial–chemical patterns while preserving the initial geometric backbone. For peptides with available conformer ensembles, the model further extracts residue-level root-mean-square fluctuation (RMSF), pairwise distance variance, contact occupancy under multiple distance cutoffs, and peptide-level root-mean-square deviation (RMSD) and radius of gyration. These statistics are incorporated through an availability-masked residual branch to refine the structural expert's output. When conformer-ensemble information is unavailable, this residual contribution is constrained to exactly zero.

Within the structural expert, the contextualized sequence and confidence-aware spatial representations are integrated through pLDDT-aware cross-attention, followed by peptide-level confidence gating; the detailed construction is given in Section 2.4.6. After masked mean pooling, fusion with global peptide-level physicochemical features, and multilayer perceptron (MLP)

classification, the sequence-enhanced structural expert obtains a base structural logit. The base structural logit is further corrected by an availability-masked, zero-initialized conformer residual before being transformed into the structural-expert probability p_{SE3D}.

2.2.3 Hierarchical Fusion and Toxicity Decision

At the top level of the model, the calibrated 1D sequence expert and the sequence-enhanced 3D structural expert are not combined through further intermediate-feature concatenation; instead, their outputs are fused with fixed weights in the probability space. The final toxicity probability is defined as

$$p_{\mathrm{GHXTox}} = 0.62\, p_{\mathrm{1D}} + 0.38\, p_{\mathrm{SE3D}}. \tag{1}$$

The higher weight assigned to the sequence expert reflects its more stable baseline discriminative ability, while the structural expert provides complementary evidence on spatial geometry, local chemical environments, and conformational dynamics. The fusion weights are determined exclusively from group-aware training out-of-fold (OOF) predictions and are frozen before evaluation on the independent test sets. In the implementation, the two expert probabilities are likewise combined linearly, without re-optimizing the fusion parameters on the test data.

Finally, the fused probability is converted into a binary toxicity prediction using a fixed decision threshold:

$$\hat{y} = \begin{cases} 1, & p_{\mathrm{GHXTox}} \geq 0.4702, \\ 0, & p_{\mathrm{GHXTox}} < 0.4702, \end{cases} \tag{2}$$

where $\hat{y} = 1$ denotes a toxic peptide and $\hat{y} = 0$ denotes a non-toxic peptide. The probability-calibration parameters, expert-fusion weights, and classification threshold are all determined solely from training out-of-fold (OOF) predictions; Test1 and Test2 are reserved exclusively for final independent evaluation. This hierarchical design enables fine-grained sequence–structure interaction through pLDDT-aware cross-attention inside the structural expert, while the frozen probability-level fusion at the top level enhances inter-expert complementarity and predictive stability.

2.3 Task-Adaptive 1D Sequence Modeling

The 1D sequence expert is designed to learn toxicity-related contextual semantics and sequential dependencies directly from peptide amino acid sequences. Instead of reducing a frozen protein-language-model output to a fixed mean vector, GHXTox preserves the residue-level representations generated by ProtT5 and adapts the pretrained encoder to the peptide toxicity prediction task through low-rank adaptation (LoRA). The resulting residue-level representations are then processed by a bidirectional Mamba state-space encoder to jointly model local residue patterns, N-to-C dependencies, and C-to-N dependencies. The encoded residue features are aggregated via masked mean pooling and passed to a linear classification head to produce the single-model toxicity probability. Finally, three models trained on the full training set with different random seeds are combined by equal-weight probability averaging, followed by Platt calibration fitted exclusively on training out-of-fold (OOF) predictions, yielding the final 1D expert probability p_{1D}.

This design organizes pretrained protein semantics, parameter-efficient task adaptation, bidirectional sequence modeling, and probability calibration into a coherent processing pipeline. While most of the pretrained knowledge in ProtT5 is retained, LoRA enables task-specific adaptation of the residue representations. The bidirectional Mamba encoder further reorganizes these representations according to toxicity-related sequential patterns, whereas multi-seed ensembling and out-of-fold calibration reduce training variance and align the calibrated probability scale with that of the downstream structural expert.

2.3.1 Residue Tokenization and LoRA-Adapted ProtT5

Given a peptide sequence of length L, we first convert its amino acid residues into the space-separated format required by the ProtT5 tokenizer and map them to token indices. Special tokens are inserted automatically, and sequences of different lengths are padded to a uniform length within each batch. The maximum token length is set to 128, which is sufficient for all peptides used in this study. Under the current ProtT5 tokenizer configuration, the first L token positions correspond to the L amino acid residues, followed by the EOS token and, when required, padding tokens. Accordingly, the implementation constructs the residue-validity mask directly from the original peptide length:

$$\mathcal{S} = (a_1, a_2, \ldots, a_L), \qquad m_i = \begin{cases} 1, & 1 \le i \le L, \\ 0, & i > L. \end{cases} \tag{3}$$

where $m_i = 1$ denotes an actual amino acid position and $m_i = 0$ denotes a special or padding position; the same mask is applied in the Mamba scans and the subsequent global pooling.

The tokenized sequence is then processed by the ProtT5-XL-UniRef50 encoder [CITE: ProtT5, Elnaggar et al.] to generate contextualized residue-level representations:

$$\mathbf{H}^{\mathrm{PLM}} = \mathrm{ProtT5}_{\mathrm{LoRA}}(\mathcal{S}), \qquad \mathbf{H}^{\mathrm{PLM}} \in \mathbb{R}^{L \times 1024}. \tag{4}$$

Retaining the complete residue-level representation matrix enables the downstream encoder to preserve position-specific contextual semantics. Each residue vector thus encodes not only its amino acid identity but also its conditional context within the complete peptide sequence.

To adapt the general-purpose ProtT5 encoder to the peptide toxicity prediction task, we apply low-rank adaptation (LoRA) [CITE: LoRA, Hu et al., 2022] to the linear transformations within the encoder. For an original linear layer $\mathbf{y} = \mathbf{W}_0\mathbf{x} + \mathbf{b}$, the pretrained weight matrix $\mathbf{W}_0$ is frozen, and the task-specific update is factorized into two trainable low-rank matrices:

$$\mathbf{W} = \mathbf{W}_0 + \frac{\alpha}{r}\mathbf{B}\mathbf{A}, \qquad \mathbf{A} \in \mathbb{R}^{r \times d_{\mathrm{in}}}, \quad \mathbf{B} \in \mathbb{R}^{d_{\mathrm{out}} \times r}. \tag{5}$$

The LoRA-augmented forward pass is therefore

$$\mathbf{y} = \mathbf{W}_0\mathbf{x} + \mathbf{b} + \frac{\alpha}{r}\mathbf{B}\mathbf{A}\,\mathrm{Dropout}(\mathbf{x}). \tag{6}$$

We set the LoRA rank to $r = 8$, the scaling coefficient to $\alpha = 16$, and the LoRA dropout rate to 0.1, yielding a scaling factor of $\alpha/r = 2$. Matrix $\mathbf{A}$ is initialized with Kaiming initialization, whereas $\mathbf{B}$ is initialized to zero; consequently, the LoRA branch initially leaves the pretrained transformation unchanged and progressively learns toxicity-specific updates during training.

Unlike conventional configurations that apply LoRA only to selected attention projections, the current implementation augments all linear layers within the ProtT5 encoder. The pretrained base transformations and embeddings remain frozen, whereas the low-rank parameters and the downstream sequence- classification modules are trainable.

The 1024-dimensional ProtT5 representations are then projected into the 192-dimensional hidden space used by the Mamba encoder [CITE: Mamba, Gu and Dao]:

$$\mathbf{X}^{(0)} = \mathrm{Dropout}\left(\mathrm{GELU}\left[\mathrm{LN}\left(\mathbf{H}^{\mathrm{PLM}}\mathbf{W}_p + \mathbf{b}_p\right)\right]\right), \qquad \mathbf{X}^{(0)} \in \mathbb{R}^{L \times 192}. \tag{7}$$

where LN denotes layer normalization. The projection block consists of a linear transformation, layer normalization, GELU activation, and dropout. It reduces the computational dimension for the downstream state-space encoder while providing a task-specific nonlinear transformation of the ProtT5 residue representations.

2.3.2 Bidirectional Mamba Sequence Encoder

Although ProtT5 provides contextualized residue representations, these representations primarily encode general protein semantics learned during large-scale pretraining. To further model the sequential dependencies specifically associated with peptide toxicity, two bidirectional Mamba layers are placed after the ProtT5 projection block. Mamba employs an input-dependent selective state-space mechanism that dynamically controls which residue information is retained, updated, or suppressed during sequential scanning [CITE: Mamba, Gu and Dao].

For a unidirectional Mamba scan, the input representation $\mathbf{X} \in \mathbb{R}^{L \times d_{\text{model}}}$ with $d_{\text{model}} = 192$ is first projected into a content branch and a gating branch:

$$[\mathbf{U}, \mathbf{Z}] = \mathbf{X}\mathbf{W}_{\text{in}} + \mathbf{b}_{\text{in}}, \qquad \mathbf{U}, \mathbf{Z} \in \mathbb{R}^{L \times d_{\text{inner}}}, \quad d_{\text{inner}} = 2d_{\text{model}} = 384, \tag{8}$$

i.e., the projection expands the feature dimension from 192 to 768 and splits it into two branches, each of dimension $d_{\text{inner}} = 384$. The content branch is processed by a depthwise one-dimensional convolution with a kernel width of 4, followed by a SiLU activation:

$$\widetilde{\mathbf{U}} = \text{SiLU}(\text{DWConv1D}(\mathbf{U})), \qquad \widetilde{\mathbf{U}}^{\text{m}}_t = m_t \widetilde{\mathbf{U}}_t, \tag{9}$$

where $\widetilde{\mathbf{U}}^{\text{m}}_t$ denotes the convolved feature after applying the residue-validity mask. This local convolution extracts short-range residue patterns before the state-space scan, enabling the module to combine local sequence motifs with the longer-range context accumulated through state recurrence. Input-dependent projections then generate the discretization step size and the state-space parameters from the masked feature:

$$[\delta_t, \mathbf{B}_t, \mathbf{C}_t] = \text{Linear}(\widetilde{\mathbf{U}}^{\text{m}}_t), \qquad \Delta_t = \text{Softplus}(\mathbf{W}_\Delta \delta_t + \mathbf{b}_\Delta). \tag{10}$$

The state-transition matrix is parameterized to be negative and is discretized according to the input-dependent step size:

$$\mathbf{A} = -\exp(\mathbf{A}_{\log}), \qquad \bar{\mathbf{A}}_t = \exp(\Delta_t \mathbf{A}), \qquad \bar{\mathbf{B}}_t = \Delta_t \mathbf{B}_t, \tag{11}$$

where $\mathbf{A}_{\log} \in \mathbb{R}^{d_{\text{inner}} \times d_{\text{state}}}$ is a learnable parameter matrix. The selective state recurrence is expressed as

$$\mathbf{S}_t = \bar{\mathbf{A}}_t \odot \mathbf{S}_{t-1} + \bar{\mathbf{B}}_t \odot \widetilde{\mathbf{U}}^{\text{m}}_t, \qquad Y_t = \sum_{n=1}^{d_{\text{state}}} S_{t,n} C_{t,n}, \tag{12}$$

where the state dimension is set to $d_{\text{state}} = 16$. The state-space output at position t is subsequently combined with the gating branch and the skip term:

$$\mathbf{O}_t = \mathbf{W}_{\text{out}} \left[\mathbf{Y}_t \odot \text{SiLU}(\mathbf{Z}_t) + \mathbf{D} \odot \widetilde{\mathbf{U}}^{\text{m}}_t \right]. \tag{13}$$

At invalid positions, $\widetilde{\mathbf{U}}^{\text{m}}_t$ is zero and Δ_t is also zeroed, preventing padding tokens from injecting information into the recurrent states.

A unidirectional scan accesses only one side of the sequence context, whereas peptide toxicity may depend on residue patterns associated with both termini and with interactions spanning different sequence positions. Therefore, in the ℓ-th bidirectional Mamba layer, forward and backward scans are applied to the normalized sequence using the residue-validity mask $\mathbf{m}$:

$$\begin{aligned} \mathbf{F}^{(\ell)} &= \mathcal{M}^{(\ell)}_f(\text{LN}(\mathbf{X}^{(\ell)}), \mathbf{m}), \\ \mathbf{R}^{(\ell)} &= \text{Reverse}\left[\mathcal{M}^{(\ell)}_b \left(\text{Reverse}\left[\text{LN}(\mathbf{X}^{(\ell)}) \right], \text{Reverse}(\mathbf{m}) \right) \right], \end{aligned} \tag{14}$$

where the backward-scan output is restored to the original residue order. The two directional outputs, each of dimension 192, are concatenated along the feature dimension (384 in total) and fused by a linear layer that projects the representation back to 192 dimensions:

$$\mathbf{M}^{(\ell)} = \text{GELU}\left(\text{Dropout}\left(\mathbf{W}_m \left[\mathbf{F}^{(\ell)} \parallel \mathbf{R}^{(\ell)} \right] + \mathbf{b}_m \right) \right), \qquad \mathbf{X}^{(\ell+1)} = \mathbf{X}^{(\ell)} + \text{Dropout}(\mathbf{M}^{(\ell)}), \tag{15}$$

where $\|$ denotes feature concatenation, the dropout inside the GELU block is applied within the fusion module, and the second dropout is applied before the residual addition. Two bidirectional Mamba layers are stacked in the final model. The first layer reconstructs bidirectional residue dependencies from the task-adapted ProtT5 features, whereas the second layer further aggregates the resulting contextual representations. This design allows each residue position to receive conditional information from both its N-terminal and C-terminal contexts.

2.3.3 Masked Pooling and Class-Weighted Classification

After two bidirectional Mamba layers, the model produces residue-level sequence representations:

$$\mathbf{X}^{(2)} = [\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_L], \qquad \mathbf{X}^{(2)} \in \mathbb{R}^{L\times 192}. \tag{16}$$

Because peptide sequences have variable lengths, GHXTox does not rely on a fixed-position classification token. Instead, masked mean pooling is applied over the valid amino acid positions:

$$\mathbf{h}_{1D} = \frac{\sum_{i=1}^{L} m_i \mathbf{x}_i}{\sum_{i=1}^{L} m_i}, \tag{17}$$

where the denominator equals the number of valid residue positions. This operation converts the variable-length residue sequence into a fixed 192-dimensional peptide-level representation while ensuring that EOS and padding positions do not contribute to the pooled feature. In contrast to global max pooling, masked mean pooling aggregates distributed evidence across the entire peptide and avoids a representation dominated by a single extreme activation.

The peptide-level representation is first normalized and then passed through a single-output linear classification head:

$$z_{1D} = \mathbf{w}_c^{\top}\,\mathrm{LN}(\mathbf{h}_{1D}) + b_c, \qquad p_{1D}^{\mathrm{raw}} = \sigma(z_{1D}) = \frac{1}{1+\exp(-z_{1D})}, \tag{18}$$

where z_{1D} denotes the 1D expert logit and p_{1D}^{raw} the corresponding uncalibrated toxicity probability. The classification head of the current 1D expert consists of layer normalization followed by a single linear layer, rather than a multilayer perceptron; it is therefore referred to as a linear classification head throughout the architecture and Methods sections.

Because non-toxic peptides substantially outnumber toxic peptides in the training data, a positive-class-weighted binary cross-entropy loss is used in the current 1D expert instead of SMOTE or Borderline-SMOTE. For a training partition containing N_+ toxic and N_- non-toxic peptides, the positive-class weight is defined as

$$w_+ = \frac{N_-}{N_+}, \tag{19}$$

and the class-weighted objective is expressed as

$$\mathcal{L}_{\mathrm{WBCE}} = -\frac{1}{N}\sum_{j=1}^{N}\left[w_+ y_j \log\sigma(z_j) + (1-y_j)\log(1-\sigma(z_j))\right]. \tag{20}$$

During five-fold out-of-fold training, w_+ is calculated independently from the class counts of each fold's training partition; for full-data training, it is recalculated using the complete training set. This strategy increases the contribution of the minority toxic class without generating synthetic peptide sequences or feature vectors, while preventing any information from the held-out validation fold from entering the training procedure.

2.3.4 Three-Seed Ensemble and OOF Platt Calibration

The predictions of deep neural networks may vary because of parameter initialization, mini-batch ordering, and stochastic optimization. To reduce the variance associated with an individual training run, three 1D expert models are trained under identical data, architecture, and optimization settings using the random seeds 42, 123, and 2025. The deployed ensemble consists of three models trained on the complete training set, rather than the fold-specific checkpoints produced during cross-validation.

For sample j, the toxicity probability produced by the model trained with seed k is denoted by $p_j^{(k)} = \sigma(z_j^{(k)})$, $k \in \{1, 2, 3\}$, and the uncalibrated ensemble score is obtained by equal-weight arithmetic averaging:

$$s_j = \frac{1}{3} \sum_{k=1}^{3} p_j^{(k)}. \tag{21}$$

No individual seed is selected according to its performance on Test1 or Test2, and no seed-specific ensemble weight is introduced. All members share the same architecture and training protocol; the different random seeds provide averaging over a limited set of initialization and optimization stochasticities.

Although probability averaging reduces prediction variance, neural-network probabilities may remain systematically overconfident or underconfident. Moreover, the top-level GHXTox framework subsequently combines the 1D expert with the structural expert, making probability-scale compatibility particularly important. Therefore, Platt scaling [CITE: Platt 1999] is applied after the three-model probability average.

Training out-of-fold predictions are first generated using the fixed group-aware five-fold partition. Each training peptide receives an OOF probability from a model trained without the fold containing that peptide. The OOF probabilities from the three random seeds are then averaged to obtain the OOF ensemble score s_j^{OOF}. A two-parameter logistic calibrator is fitted using only these training OOF scores and their labels:

$$p_{1D,j} = \sigma\left(a \,\mathrm{logit}(s_j) + b\right), \tag{22}$$

where $\mathrm{logit}(s) = \ln\left(s/(1-s)\right)$, and a and b denote the Platt calibration parameters; the calibrator operates on the logit of the averaged probability score. The parameters are estimated as

$$(a, b) = \arg\min_{a,b} \sum_{j=1}^{N} \mathrm{BCE}\left[y_j,\ \sigma\left(a \,\mathrm{logit}\left(s_j^{\mathrm{OOF}}\right) + b\right)\right]. \tag{23}$$

The scaling parameter is reparameterized as $a = \exp(\eta) > 0$, which guarantees a monotonic probability transformation and therefore preserves the sample ranking of the ensemble score; AUROC and AUPRC are consequently unchanged by calibration. For independent test or previously unseen peptides, the output probabilities of the three full-data models are first averaged, after which the frozen calibration parameters are applied:

$$p_{1\mathrm{D}} = \sigma\left(a \,\mathrm{logit}\left(\frac{1}{3} \sum_{k=1}^{3} p^{(k)}\right) + b\right). \tag{24}$$

The labels of Test1 and Test2 are not used for ensemble-member selection, calibration fitting, or 1D threshold optimization. The calibrated probability $p_{1\mathrm{D}}$ is the sequence-expert output used in the top-level probability fusion described in Section 2.2.3. No additional Platt calibrator is fitted after the final GHXTox probability fusion. The complete order of operations is therefore

3-seed probabilities $\rightarrow$ averaging $\rightarrow$ OOF-fitted Platt calibration $\rightarrow p_{1\mathrm{D}} \rightarrow$ 1D/3D fusion.

2.4 pLDDT-Aware Multi-Scale 3D Structural Modeling

Peptide toxicity is influenced not only by amino acid composition and linear residue order but also by the three-dimensional (3D) arrangement of charged groups, hydrophobic regions, aromatic systems, and sulfur-containing moieties. However, peptides are generally short, conformationally flexible, and poorly covered by experimentally determined structures. Treating a single predicted conformation as an error-free structural observation may therefore amplify structure-prediction errors. Conversely, representing a peptide with only C_α coordinates or peptide-level summary descriptors fails to capture the local spatial-chemical environments that are relevant to toxicity.

To address these limitations, GHXTox employs a pLDDT-aware multi-scale structural expert that captures complementary information at five levels: residue semantics, spatial residue neighborhoods, full-atom chemical sites, peptide-level global properties, and conformer-ensemble statistics. The model first extracts residue-level C_α coordinates, full-atom coordinates, and normalized pLDDT confidence scores from the predicted structure. An internal sequence-semantic subbranch combines amino acid residue types, functional-group categories, explicit physicochemical descriptors, and ESM2 residue embeddings to generate contextualized sequence states. In parallel, a spatial-geometric subbranch constructs a hybrid residue graph containing both sequential adjacency edges and spatial-neighborhood edges and propagates geometric information through pLDDT-aware equivariant graph layers.

To overcome the limited chemical resolution of a C_α-only residue graph, GHXTox additionally extracts typed chemical sites from the full-atom structure and performs message passing using dual-scale distances, directional relationships, interaction-compatibility channels, and structural confidence. For peptides with available conformer ensembles, an additional statistical residual branch summarizes residue fluctuations, distance variation, and contact occupancy. The residual is constrained to exactly zero when conformer information is unavailable.

Rather than discarding low-confidence structures through hard filtering, GHXTox treats pLDDT as a continuous reliability signal and incorporates it into geometric edge weighting, chemical-site messages, residue-level structural gating, and peptide-level structural gating. Low-confidence structural evidence is therefore down-weighted rather than removed, preserving structural coverage while limiting the influence of uncertain coordinates. Importantly, pLDDT is interpreted here as a local confidence estimate produced by the structure-prediction model, rather than an experimentally determined measure of structural accuracy, conformational flexibility, or biological activity. The detailed architecture of the structural expert is illustrated in Figure 2.

Figure 2: Detailed architecture of the sequence-enhanced structural expert. The structural expert integrates ESM2-enhanced residue semantics, a pLDDT-aware geometric residue graph, full-atom chemical-site message passing, and availability-masked conformer-ensemble statistics. Residue-level structural states are continuously modulated by node-level confidence before serving as keys and values in sequence–structure cross-attention. A peptide-level confidence gate further controls the fused structural context, while conformer statistics provide a zero-initialized logit residual only when valid ensemble information is available.

2.4.1 Peptide Structure Prediction and Conformer Preparation

For a peptide sequence of length L, $S = (a_1, a_2, \ldots, a_L)$, ESMFold is first used to predict its three-dimensional structure [CITE: ESMFold]. The predicted structure is stored as a full-atom PDB file, from which residue-level C_α coordinates, backbone coordinates, full-atom-derived chemical-site coordinates, and pLDDT scores are extracted. This operation is denoted as

$$(\mathbf{X}, \mathcal{A}, \mathbf{q}) = \mathrm{ESMFold}(\mathcal{S}), \qquad \mathbf{X} = [\mathbf{x}_1, \ldots, \mathbf{x}_L]^\top \in \mathbb{R}^{L\times 3}, \quad \mathbf{q} = (q_1, \ldots, q_L) \in [0,1]^L, \tag{25}$$

where $\mathbf{X}$ contains the C_α coordinates, $\mathcal{A}$ denotes the residue-organized full-atom coordinate set, and $\mathbf{q}$ contains the residue-level pLDDT confidence scores rescaled to $[0, 1]$. The C_α coordinates serve as the primary spatial variables of the residue-level equivariant graph, whereas the full-atom coordinates are used to construct local chemical sites. Before geometric encoding, the coordinates are centered at the peptide centroid to remove the effect of absolute translation.

A single predicted structure provides a common structural anchor but cannot represent potential conformational heterogeneity. Therefore, the current default GHXTox configuration additionally uses conformer-ensemble statistics on a prespecified subset of peptides covered by PepFlow [CITE: PepFlow]. Up to eight validated and aligned conformers are retained for each available peptide, with residue indices matched to a common sequence order. The ensemble is represented as

$$\mathcal{C} = \{\mathbf{X}^{(1)}, \mathbf{X}^{(2)}, \ldots, \mathbf{X}^{(M)}\}, \qquad 1 \le M \le 8, \quad \mathbf{X}^{(m)} \in \mathbb{R}^{L\times 3}, \tag{26}$$

where $\mathbf{X}^{(m)}$ denotes the C_α coordinates of conformer m. Rigid-body alignment is performed before the ensemble statistics are calculated, ensuring that the resulting variation reflects internal geometric changes rather than arbitrary global rotations or translations.

Because conformer ensembles are not available for every training peptide, a binary availability indicator is defined as

$$c_{\text{conf}} = \begin{cases} 1, & \text{if valid ensemble statistics are available,} \\ 0, & \text{otherwise.} \end{cases} \tag{27}$$

When $c_{\text{conf}} = 0$, the conformer-statistics tensors are filled with zeros, and the corresponding residual contribution is explicitly suppressed by the availability indicator. Missing ensemble information is therefore not interpreted as evidence of structural rigidity; instead, the model falls back exactly to the single-structure pathway.

The generated conformers are computational samples rather than experimentally determined trajectories or rigorous thermodynamic ensembles. Consequently, RMSF, contact occupancy, and distance variance are interpreted as sampling-derived descriptors of conformational variation rather than direct measurements of peptide dynamics in solution.

2.4.2 Residue Feature Construction and Hybrid Geometric Graph

The structural expert constructs both an internal sequence representation and a spatial-geometric representation at the residue level. Importantly, the internal sequence-semantic subbranch described here uses frozen ESM2 residue embeddings and a lightweight Transformer. It is distinct from the LoRA-ProtT5 and bidirectional Mamba expert described in Section 2.3. The former supports residue-level sequence–structure alignment inside the structural expert, whereas the latter acts as an independent top-level 1D prediction expert.

For residue i, the input to the internal sequence-semantic subbranch is

$$\mathbf{u}_i^{\text{seq}} = \left[\mathbf{e}_i^{\text{AA}} \parallel \mathbf{e}_i^{\text{group}} \parallel \mathbf{f}_i^{\text{res}} \parallel P_{\text{ESM}}\left(\mathbf{e}_i^{\text{ESM2}}\right)\right], \tag{28}$$

where $\mathbf{e}_i^{\text{AA}} \in \mathbb{R}^{32}$ is the amino acid category embedding, $\mathbf{e}_i^{\text{group}} \in \mathbb{R}^{12}$ is the functional-group category embedding, $\mathbf{f}_i^{\text{res}} \in \mathbb{R}^{42}$ denotes the explicit residue-level physicochemical descriptors, and $\mathbf{e}_i^{\text{ESM2}} \in \mathbb{R}^{1280}$ is the ESM2-650M residue representation; the projection P_{ESM} maps the ESM2 representation to 128 dimensions. Stacking the residue-level feature vectors yields the sequence-feature matrix $U_{\text{seq}} \in \mathbb{R}^{L\times 214}$, where $214 = 32+12+42+128$. After a linear projection and positional encoding, the sequence is processed by a two-layer Transformer encoder with hidden dimension 128, four attention heads, and a feed-forward dimension of 512:

$$\mathbf{H}^{\text{seq}} = \text{Transformer}_2\left[P_{\text{seq}}\left(U_{\text{seq}}\right) + E_{\text{pos}}\right], \qquad \mathbf{H}^{\text{seq}} \in \mathbb{R}^{L\times 128}, \tag{29}$$

This representation encodes the contextual sequence semantics of each residue and subsequently serves as the query representation in the sequence–structure cross-attention module.

In the spatial subbranch, the initial representation of residue i is constructed as

$$\mathbf{h}_i^{(0)} = P_{\mathrm{sp}}\left[\mathbf{f}_i^{\mathrm{res}} \parallel \mathbf{g}_i^{\mathrm{geo}} \parallel q_i \parallel P_{\mathrm{ESM}}\left(\mathbf{e}_i^{\mathrm{ESM2}}\right)\right], \tag{30}$$

where $\mathbf{g}_i^{\mathrm{geo}} \in \mathbb{R}^{16}$ is a lightweight geometric descriptor derived from the C_α coordinates and pLDDT scores. The 16-dimensional descriptor includes (i) residue-level pLDDT and local confidence statistics; (ii) radial position, contact fractions, and soft spatial density; (iii) nearest-neighbor, nonlocal-neighbor, and local sequential distances; and (iv) sine/cosine encodings of the C_α-trace angles and torsions. These explicit descriptors provide stable local geometric priors without introducing an overly complex handcrafted representation for short and uncertain predicted peptide structures.

GHXTox then constructs a hybrid residue graph:

$$\mathcal{G} = (\mathcal{V}, \mathcal{E}_{\mathrm{seq}} \cup \mathcal{E}_{\mathrm{knn}}), \tag{31}$$

where each residue corresponds to one node. Sequential edges $\mathcal{E}_{\mathrm{seq}}$ connect adjacent residues and preserve peptide-chain continuity, whereas $\mathcal{E}_{\mathrm{knn}}$ connects each residue to its 10 nearest spatial neighbors and is symmetrized. The hybrid graph therefore captures both covalent sequence order and nonlocal spatial contacts while avoiding large variations in graph density that may arise from a single fixed-distance cutoff. Equivalently, $A_{ij} \in \{0, 1\}$ denotes whether residues i and j are connected by a sequential or k-nearest-neighbor edge in the hybrid graph.

For residues i and j, the C_α distance is

$$d_{ij} = \|\mathbf{x}_i - \mathbf{x}_j\|_2, \qquad \phi_k(d_{ij}) = \exp\left[-\gamma\,(d_{ij} - \mu_k)^2\right], \quad \gamma = \frac{1}{(\mu_{k+1} - \mu_k)^2}, \quad k = 1, \ldots, 16, \tag{32}$$

where the centers μ_k are uniformly spaced over the range 0–24 Å. The base edge representation additionally includes the pairwise structural confidence and the normalized sequence separation:

$$c_{ij} = \sqrt{q_i q_j}, \qquad s_{ij} = \frac{\log(1 + |i - j|)}{\log(L + 1)}. \tag{33}$$

Thus, sequentially adjacent residues receive small values, whereas residues separated by longer sequence intervals approach one. The resulting edge feature is therefore $\mathbf{e}_{ij} = [\boldsymbol{\phi}(d_{ij}) \parallel c_{ij} \parallel s_{ij}]$. This formulation allows the model to distinguish sequentially adjacent contacts, long-range spatial contacts, and contacts supported by different levels of structural confidence.

2.4.3 pLDDT-Aware Equivariant Graph Encoding

To learn three-dimensional representations without relying on a particular global coordinate frame, the spatial branch employs three E(3)-equivariant graph neural network layers [CITE: EGNN]. The network updates residue features using rotation- and translation-invariant quantities while updating coordinates equivariantly. Consequently, a global rigid transformation of the peptide structure does not change the predicted toxicity, whereas intermediate coordinate representations transform consistently with the input.

At layer ℓ, the message from residue j to residue i is computed as

$$\mathbf{m}_{ij}^{(\ell)} = \phi_e^{(\ell)}\left[\mathbf{h}_i^{(\ell)} \parallel \mathbf{h}_j^{(\ell)} \parallel \mathbf{e}_{ij}^{(\ell)}\right], \tag{34}$$

where $\phi_e^{(\ell)}$ is a message multilayer perceptron. To prevent uncertain or distant contacts from contributing as strongly as local high-confidence contacts, a continuous edge weight is defined as

$$w_{ij} = A_{ij}\,\exp\!\left(-\frac{d_{ij}}{12}\right)\left[q_{\min} + (1 - q_{\min})\,c_{ij}^{\rho}\right], \qquad q_{\min} = 0.1, \quad \rho = 1.5, \tag{35}$$

The term $\exp(-d_{ij}/12)$ imposes a smooth distance decay with a characteristic decay length of 12 Å. Here, $A_{ij} \in \{0, 1\}$ is the adjacency indicator of the hybrid graph defined in Section 2.4.2, and c_{ij} is the pairwise structural confidence defined in Section 2.4.2. The minimum weight $q_{\min}$ prevents low-confidence regions from being completely disconnected, whereas the exponent ρ increases the relative contribution of high-confidence contacts. The distance-decay term further reduces the influence of more distant spatial neighbors.

Messages received by node i are aggregated with a normalization floor:

$$\mathbf{m}_i^{(\ell)} = \frac{\sum_j w_{ij}\, \mathbf{m}_{ij}^{(\ell)}}{\max\Big(\sum_j w_{ij},\, 1\Big)}. \tag{36}$$

The aggregated message is used to update the node representation with a residual connection followed by layer normalization:

$$\mathbf{h}_i^{(\ell+1)} = \mathrm{LN}\left\{\mathbf{h}_i^{(\ell)} + \phi_h^{(\ell)}\left[\mathbf{h}_i^{(\ell)} \parallel \mathbf{m}_i^{(\ell)}\right]\right\}. \tag{37}$$

In addition to updating node features, the EGNN refines residue coordinates using pairwise relative vectors. A bounded scalar coordinate-update coefficient is generated from the edge message, and the coordinate update is

$$\begin{aligned} \alpha_{ij}^{(\ell)} &= \tanh\left[\phi_x^{(\ell)}\left(\mathbf{m}_{ij}^{(\ell)}\right)\right] w_{ij}, \\ \mathbf{x}_i^{(\ell+1)} &= \mathbf{x}_i^{(\ell)} + 0.05\, \frac{\sum_j \alpha_{ij}^{(\ell)}\left(\mathbf{x}_i^{(\ell)} - \mathbf{x}_j^{(\ell)}\right)}{\max\Big(\sum_j w_{ij},\, 1\Big)}. \end{aligned} \tag{38}$$

The coordinates are recentered after every layer to reduce numerical drift. Because the update depends only on scalar messages, distances, and relative coordinate vectors, the operation remains equivariant to rigid rotations and translations.

Three EGNN layers are stacked to progressively enlarge the spatial receptive field. The first layer primarily encodes direct geometric neighbors, whereas subsequent layers integrate higher-order spatial environments through repeated message passing. Unlike approaches that simply append pLDDT as an ordinary node feature, GHXTox uses pLDDT to directly modulate the strength of geometric messages, allowing structural uncertainty to influence representation learning at the propagation stage.

Continuous down-weighting does not imply that a low-pLDDT residue is biologically unimportant. Low confidence may arise from prediction error or potential conformational flexibility; the model reduces the reliability assigned to uncertain coordinate evidence while preserving the residue's sequence-semantic and physicochemical features.

2.4.4 Full-Atom Chemical-Site Message Passing

A C_α-based residue graph describes the global fold and spatial neighborhoods but cannot distinguish chemically distinct regions within the same side chain. For example, tyrosine contains both an aromatic ring and a hydroxyl group, whereas tryptophan contains an aromatic system and an indole nitrogen. Compressing each side chain into a single centroid would mix these chemically different moieties. GHXTox therefore extracts chemistry-aware pseudo-sites from the predicted full-atom PDB structure and processes them with a dedicated local chemical branch.

Up to two chemical sites are retained for each residue. Site identity is represented as an eight-dimensional multi-hot vector allowing overlapping types:

$$\mathbf{t}_{ir} \in \{0, 1\}^8, \tag{39}$$

corresponding to positive, negative, donor, acceptor, aromatic, hydrophobic, sulfur, and conformational, where r indexes the chemical site within residue i. Multiple types may be assigned to the same site. For example, the cysteine sulfur site is labeled as donor, hydrophobic, and sulfur-containing, whereas the proline ring is labeled as hydrophobic and conformationally constrained. The aromatic ring and hydroxyl group of tyrosine are represented as separate sites, as are the aromatic system and the indole NE1 nitrogen (donor) of tryptophan.

A site coordinate is obtained from the mean position of the heavy atoms defining the corresponding chemical moiety. Site orientation is represented by either a bond direction or an unsigned plane normal. Because the sign of a plane normal is arbitrary, downstream directional features use absolute cosines, preserving rotation invariance and eliminating orientation-sign ambiguity.

The initial representation of site r in residue i is

$$\mathbf{z}_{ir}^{(0)} = \phi_{\text{site}}\left[\mathbf{t}_{ir} \parallel q_i \parallel o_{ir}^{\text{valid}}\right], \tag{40}$$

where q_i is the pLDDT score of the parent residue and o_{ir}^{valid} indicates whether a reliable orientation could be constructed; the input is projected into a 64-dimensional hidden space.

Chemical-site edges are established between valid sites separated by no more than 16 Å. In the current configuration, same-residue site edges are retained, preserving spatial relationships between chemically different moieties in complex side chains. Each edge contains four complementary feature families. First, the raw distance in Å is encoded using 16 radial basis functions, $\boldsymbol{\phi}_{\text{raw}}(d_{uv}) \in \mathbb{R}^{16}$. Second, a peptide-size-normalized distance is constructed. The radius of gyration of the C_α coordinates is

$$R_g = \sqrt{\frac{1}{L}\sum_{i=1}^{L} \|\mathbf{x}_i - \bar{\mathbf{x}}\|_2^2}, \qquad \bar{\mathbf{x}} = \frac{1}{L}\sum_{i=1}^{L}\mathbf{x}_i, \qquad \bar{d}_{uv} = \frac{d_{uv}}{R_g}, \tag{41}$$

where $\bar{\mathbf{x}}$ denotes the centroid of the valid C_α coordinates, and the normalized distance is encoded by eight radial basis functions over the interval 0–4, $\boldsymbol{\phi}_{\text{norm}}(\bar{d}_{uv}) \in \mathbb{R}^8$. The raw representation describes the physical separation in Å, whereas the normalized representation describes the separation relative to the overall peptide size. Their combination allows the model to distinguish, for example, an 8 Å contact in a compact short peptide from an 8 Å contact in a larger peptide.

Third, four rotation-invariant orientation features are used:

$$\mathbf{o}_{uv} = \left[|\mathbf{o}_u \cdot \mathbf{r}_{uv}|,\ |\mathbf{o}_v \cdot \mathbf{r}_{vu}|,\ |\mathbf{o}_u \cdot \mathbf{o}_v|,\ o_{uv}^{\text{valid}}\right], \tag{42}$$

which describe the alignment of the source orientation with the intersite direction, the alignment of the target orientation with the reverse direction, the mutual orientation of the two sites, and orientation availability. Fourth, six explicit compatibility channels are defined:

$$\mathbf{c}_{uv}^{\text{chem}} = [c_{\text{opp}}, c_{\text{DA}}, c_{\text{aro}}, c_{\text{cat-}\pi}, c_{\text{SS}}, c_{\text{hyd}}], \tag{43}$$

representing opposite-charge, donor–acceptor, aromatic–aromatic, positive–aromatic (cation–π\$), sulfur–sulfur, and hydrophobic–hydrophobic compatibility. These channels encode candidate chemical compatibility patterns available to the model; they do not imply that a true salt bridge, hydrogen bond, cation–π\$ interaction, disulfide bond, or hydrophobic core is present in the predicted structure.

The full chemical-site edge representation is

$$\mathbf{e}_{uv}^{\text{chem}} = \left[\boldsymbol{\phi}_{\text{raw}}(d_{uv}) \parallel \boldsymbol{\phi}_{\text{norm}}(\bar{d}_{uv}) \parallel \mathbf{o}_{uv} \parallel \mathbf{c}_{uv}^{\text{chem}} \parallel \sqrt{q_u q_v}\right]. \tag{44}$$

At layer l, the site message, its pairwise-confidence-weighted form, and the confidence-normalized aggregation are

$$\begin{aligned}
\mathbf{m}_{uv}^{(l)} &= \phi_{\text{chem}}^{(l)}\left[\mathbf{z}_u^{(l)} \parallel \mathbf{z}_v^{(l)} \parallel \mathbf{e}_{uv}^{\text{chem}}\right], \\
\tilde{\mathbf{m}}_{uv}^{(l)} &= \sqrt{q_u q_v}\,\mathbf{m}_{uv}^{(l)}, \\
\bar{\mathbf{m}}_v^{(l)} &= \frac{\sum_{u\in\mathcal{N}(v)} \tilde{\mathbf{m}}_{uv}^{(l)}}{\max\left(\sum_{u\in\mathcal{N}(v)} \sqrt{q_u q_v},\, 1\right)}.
\end{aligned} \tag{45}$$

After two message-passing layers, valid sites belonging to the same residue are mean-pooled, and a residue-level chemical residual is obtained:

$$\begin{aligned}
\Delta\mathbf{h}_i^{\text{chem}} &= \mathbf{W}_{\text{chem}} \operatorname*{Mean}_{r\in\mathcal{S}_i}\left(\mathbf{z}_{ir}^{(2)}\right), \\
\mathbf{W}_{\text{chem}}^{(0)} &= 0, \quad \mathbf{b}_{\text{chem}}^{(0)} = 0, \\
\mathbf{h}_i^{\text{sp}} &= \mathbf{h}_i^{\text{EGNN}} + \Delta\mathbf{h}_i^{\text{chem}},
\end{aligned} \tag{46}$$

where the projection $\mathbf{W}_{\text{chem}}$ and its bias $\mathbf{b}_{\text{chem}}$ are initialized to zero, so that $\Delta\mathbf{h}_i^{\text{chem}} = 0$ at the beginning of training. This conservative residual design allows the new chemical-site branch to progressively learn useful corrections from an intact geometric backbone rather than perturbing the existing spatial representation with a randomly initialized contribution.

2.4.5 Availability-Masked Conformer-Ensemble Residual

A single predicted structure provides a deterministic spatial anchor, whereas a peptide may adopt multiple related conformations. Running the complete geometric network independently on every conformer would be computationally expensive and could allow the model to exploit differences between conformer generators or uneven ensemble coverage. The current GHXTox model therefore uses a lightweight conformer-statistics residual. The single-structure expert remains the base predictor, and ensemble statistics are permitted to correct its logit only when valid conformer information is available.

For a peptide with M aligned conformers, the residue-level root-mean-square fluctuation is

$$\text{RMSF}_i = \sqrt{\frac{1}{M}\sum_{m=1}^{M}\left\|\mathbf{x}_i^{(m)} - \bar{\mathbf{x}}_i\right\|_2^2}, \qquad \bar{\mathbf{x}}_i = \frac{1}{M}\sum_{m=1}^{M}\mathbf{x}_i^{(m)}. \tag{47}$$

For each residue pair i, j, the mean distance and the distance standard deviation are

$$\bar{d}_{ij} = \frac{1}{M}\sum_{m=1}^{M} d_{ij}^{(m)}, \qquad \sigma_{ij}^{d} = \sqrt{\frac{1}{M}\sum_{m=1}^{M}\left(d_{ij}^{(m)} - \bar{d}_{ij}\right)^2}. \tag{48}$$

For a distance cutoff $r \in \{6, 8, 10\}$ Å, the contact occupancy and the 8 Å contact-occupancy uncertainty are

$$O_{ij}^{(r)} = \frac{1}{M}\sum_{m=1}^{M}\mathbb{I}\left(d_{ij}^{(m)} < r\right), \qquad U_{ij}^{(8)} = O_{ij}^{(8)}\left(1 - O_{ij}^{(8)}\right). \tag{49}$$

Values close to zero or one indicate contacts that are consistently absent or present across the sampled conformers, whereas values near 0.5 indicate contact switching.

The ensemble statistics are organized into an eight-dimensional residue-level descriptor $\mathbf{r}_i^{\text{conf}} \in \mathbb{R}^8$, containing the residue RMSF; the mean pairwise distance to other residues; the standard deviation of the mean pairwise distances associated with the residue; the mean pairwise-distance standard deviation; the mean contact occupancies at 6, 8, and 10 Å; and the mean 8 Å

contact-occupancy uncertainty, which together form the eight-dimensional descriptor. A twelve-dimensional peptide-level descriptor $\mathbf{g}^{\text{conf}} \in \mathbb{R}^{12}$ is also constructed, containing the normalized number of conformers; the mean, standard deviation, and maximum of RMSF; the mean, standard deviation, and maximum of the conformer RMSD; the mean and standard deviation of the radius of gyration; the mean and maximum of the pair-distance standard deviation; and the global mean of $U_{ij}^{(8)}$, which together form the twelve-dimensional descriptor. Before neural encoding, distance- and fluctuation-based statistics are rescaled by fixed constants to place heterogeneous descriptors on comparable numerical ranges; the exact scaling factors are reported in the implementation-details table.

The residue-level ensemble features are encoded and aggregated using both masked mean and masked maximum pooling, whereas the global descriptor is processed by a separate encoder:

$$\mathbf{h}_{\text{conf}}^{\text{res}} = \left[\text{MeanPool}\left(\phi_r\left(\mathbf{R}^{\text{conf}}\right)\right) \parallel \text{MaxPool}\left(\phi_r\left(\mathbf{R}^{\text{conf}}\right)\right)\right], \qquad \mathbf{h}_{\text{conf}}^{\text{global}} = \phi_g\left(\mathbf{g}^{\text{conf}}\right). \tag{50}$$

The base structural embedding, the residue-level ensemble embedding, and the global ensemble embedding are concatenated to produce a raw logit residual:

$$\Delta z_{\text{conf}}^{\text{raw}} = \phi_{\text{conf}}\left[\mathbf{h}_{\text{base}} \parallel \mathbf{h}_{\text{conf}}^{\text{res}} \parallel \mathbf{h}_{\text{conf}}^{\text{global}}\right]. \tag{51}$$

The final linear layer of the conformer residual head is initialized to zero. Using the availability indicator c_{conf} defined in Eq. (27), the learnable conformer weight, the effective residual, and the final structural-expert logit are

$$\begin{aligned} \lambda_{\text{conf}} &= \sigma\left(\beta_{\text{conf}}\right), \qquad \lambda_{\text{conf}}^{(0)} = 0.25, \\ \Delta z_{\text{conf}} &= c_{\text{conf}}\, \lambda_{\text{conf}}\, \Delta z_{\text{conf}}^{\text{raw}}, \\ z_{\text{SE3D}} &= z_{\text{base}} + \Delta z_{\text{conf}}, \qquad c_{\text{conf}} = 0 \;\Rightarrow\; z_{\text{SE3D}} = z_{\text{base}}. \end{aligned} \tag{52}$$

where λ_{conf} is initialized to 0.25 at the beginning of training. Zero-filled feature tensors alone would not guarantee a zero correction because the residual network may contain nonzero biases; the additional hard availability mask ensures an exact fallback to the single-structure prediction. The conformer branch is permitted only to refine the single-structure predictor rather than replace the geometric backbone, reducing the risk of overfitting to limited and uneven ensemble coverage.

2.4.6 pLDDT-Aware Sequence–Structure Cross-Attention

After the EGNN and the chemical-site residual are applied, the model obtains the chemical-enhanced spatial representation

$$\mathbf{H}^{\text{sp}} = [\mathbf{h}_1^{\text{sp}}, \ldots, \mathbf{h}_L^{\text{sp}}]^\top \in \mathbb{R}^{L \times 128}, \tag{53}$$

where $\mathbf{h}_i^{\text{sp}}$ is defined in Eq. (46). Spatial information alone does not determine whether a structural pattern is relevant to the current sequence context. For example, similar hydrophobic contacts may have different implications depending on residue identities and sequence positions. GHXTox therefore uses pLDDT-aware cross-attention within the structural expert, allowing contextualized sequence states to selectively retrieve spatial evidence.

A node-level gate is first constructed for each residue. The base gate uses the 42-dimensional residue descriptors and the pLDDT score:

$$\begin{aligned} g_i^{\text{base}} &= \phi_{\text{node}}\left[\mathbf{f}_i^{\text{res}} \parallel q_i\right], \\ \mathbf{c}_i^{\text{node}} &= \left[\frac{q_i - \mu_q}{T_q},\ \frac{(q_i - \mu_q)^2}{T_q^2},\ \mathbb{I}(q_i < 0.55)\right], \\ \mu_q &= 0.70, \quad T_q = 0.12, \end{aligned} \tag{54}$$

where the residual confidence vector $\mathbf{c}_i^{\text{node}}$ increases the sensitivity of the gate to the distribution of structural confidence. The residual scale and the final node gate are

$$r_i^{\text{node}} = 0.5 + \sigma\left(\mathbf{w}_n^\top \mathbf{c}_i^{\text{node}} + b_n\right), \qquad g_i^{\text{node}} = \text{clamp}\left(g_i^{\text{base}}\, r_i^{\text{node}},\ 0, 1\right), \tag{55}$$

and the spatial representation is continuously down-weighted as

$$\widehat{\mathbf{H}}^{\text{sp}} = \mathbf{H}^{\text{sp}} \odot \mathbf{g}^{\text{node}}, \qquad \mathbf{g}^{\text{node}} = \left[g_1^{\text{node}}, \ldots, g_L^{\text{node}}\right]^\top \in [0,1]^L, \tag{56}$$

where the node-level gate vector $\mathbf{g}^{\text{node}}$ is broadcast across the feature dimension and each gate value is clamped to $[0, 1]$.

In the cross-attention operation, the contextualized sequence representation serves as the query, and the gated spatial representation serves as the key and value:

$$\mathbf{H}^{\text{cross}} = \text{MHA}\left(Q = \mathbf{H}^{\text{seq}}, K = \widehat{\mathbf{H}}^{\text{sp}}, V = \widehat{\mathbf{H}}^{\text{sp}}\right). \tag{57}$$

This direction has a direct interpretation: each residue position retrieves spatial evidence according to its contextual sequence representation. The operation is not a direct feature interaction between the LoRA-ProtT5–Mamba expert and the structural expert; it occurs internally between the ESM2-enhanced sequence states and the geometric states of the structural expert.

In addition to residue-level gating, a peptide-level gate is generated from the overall structure-confidence profile. The model first computes

$$\bar{q} = \frac{1}{L}\sum_{i=1}^{L} q_i, \qquad q_{\min} = \min_i q_i, \qquad r_{\text{low}} = \frac{1}{L}\sum_{i=1}^{L} \mathbb{I}(q_i < 0.55), \tag{58}$$

and constructs the peptide-level confidence vector $\mathbf{c}^{\text{pep}} = [(\bar{q} - \mu_q)/T_q,\ (q_{\min} - \mu_q)/T_q,\ r_{\text{low}}]$. The base peptide-level gate and its residual-adjusted form are

$$\begin{aligned} g_{\text{pep}}^{\text{base}} &= \phi_{\text{global}}\left[\bar{q},\ q_{\min},\ r_{\text{low}}\right], \\ r_{\text{pep}} &= 0.5 + \sigma\left(\mathbf{w}_{\text{pep}}^\top \mathbf{c}^{\text{pep}} + b_{\text{pep}}\right), \\ g_{\text{pep}} &= \text{clamp}\left(g_{\text{pep}}^{\text{base}}\, r_{\text{pep}},\ 0, 1\right), \end{aligned} \tag{59}$$

where the peptide-level residual scale is analogous to, but separately parameterized from, the node-level residual scale, and $g_{\text{pep}} \in [0, 1]$ denotes the scalar peptide-level structural gate.

The final residue-level fusion representation is

$$\mathbf{H}^{\text{fused}} = \text{LN}\left[\mathbf{H}^{\text{seq}} \parallel g_{\text{pep}}\, \mathbf{H}^{\text{cross}}\right]. \tag{60}$$

The two gating levels answer complementary questions: how reliable the structural evidence is at an individual residue position, and how strongly the structural branch should be trusted for the peptide as a whole. The sequence representation is always retained, whereas the structural context is dynamically scaled according to prediction confidence. Consequently, when the predicted structure has low overall confidence, the structural expert can still rely on its internal sequence-semantic pathway without allowing uncertain coordinates to dominate the classification.

2.4.7 Global Feature Integration and Structural Classification

After pLDDT-aware cross-attention, each valid residue is represented by a 256-dimensional fused vector obtained by concatenating the 128-dimensional internal sequence state with the 128-dimensional structural context:

$$\mathbf{H}^{\text{fused}} \in \mathbb{R}^{L \times 2d_h}, \qquad d_h = 128, \tag{61}$$

so that $\mathbf{H}^{\text{fused}} \in \mathbb{R}^{L\times 256}$. Masked mean pooling is applied over the valid residue positions:

$$\mathbf{h}_{\text{pool}} = \frac{\sum_{i=1}^{L} m_i \, \mathbf{H}_i^{\text{fused}}}{\sum_{i=1}^{L} m_i} \in \mathbb{R}^{256}, \tag{62}$$

where the denominator equals the number of valid residue positions and the mask excludes padding positions from the peptide representation. In contrast to maximum pooling, mean pooling integrates structural evidence distributed across multiple residue positions, which is appropriate when toxicity-related spatial patterns are not confined to a single local region.

The residue graph primarily models local and intermediate-range environments, whereas a finite number of message-passing layers may not explicitly recover all peptide-level physicochemical statistics. GHXTox therefore includes five global physicochemical descriptors, normalized to comparable scales before concatenation:

$$\mathbf{f}_{\text{pep}}^{\text{phys}} = \left[\frac{L}{128},\ \frac{Q_{\text{net}}}{L},\ f_{\text{aro}},\ f_{\text{Cys}},\ \frac{\bar{h}_{\text{hyd}}}{4.5} \right] \in \mathbb{R}^5, \tag{63}$$

representing the normalized peptide length, the length-normalized net charge, the aromatic-residue fraction, the cysteine fraction, and the scaled mean hydropathy. The classifier input is

$$\mathbf{h}_{\text{cls}} = \left[\mathbf{h}_{\text{pool}} \parallel \mathbf{f}_{\text{pep}}^{\text{phys}} \right] \in \mathbb{R}^{261}. \tag{64}$$

The base structural logit is produced by a three-layer multilayer perceptron, denoted compactly as

$$z_{\text{base}} = \phi_{\text{cls}} \left(\mathbf{h}_{\text{cls}} \right), \tag{65}$$

with dimensions $261 \to 128 \to 64 \to 1$, where each hidden layer applies GELU activation and dropout.

Applying the availability-masked conformer correction defined in Eq. (52) yields the final structural-expert logit, and the structural-expert probability is

$$z_{\text{SE3D}} = z_{\text{base}} + c_{\text{conf}} \, \lambda_{\text{conf}} \, \Delta z_{\text{conf}}^{\text{raw}}, \qquad p_{\text{SE3D}} = \sigma \left(z_{\text{SE3D}} \right),$$

as defined in Eq. (52). No standalone structural threshold is used to generate the final GHXTox class at this stage. Instead, p_{SE3D} is passed directly to the top-level dual-expert fusion described in Section 2.2.3, where it is combined with the calibrated 1D expert probability p_{1D} using frozen fusion weights. A standalone structural threshold is used only for structural-branch ablations or independent expert evaluation and does not determine the final GHXTox prediction.

The structural expert can therefore be summarized as

$$\begin{aligned} &\text{sequence-enhanced residue semantics} + \text{confidence-aware geometry} \\ &+ \text{full-atom chemical sites} + \text{global physicochemical descriptors} \\ &+ \text{availability-masked conformer statistics} \longrightarrow p_{\text{SE3D}}. \end{aligned}$$

The central contribution is not merely the inclusion of additional structural features, but the explicit separation of structural information by spatial scale and reliability. The geometric graph captures global residue relationships, the chemical-site branch resolves local functional moieties, the conformer residual summarizes sampled structural variation, and pLDDT continuously modulates edge propagation, chemical messages, and sequence–structure fusion. This design allows the structural expert to incorporate complementary 3D evidence while handling structure-prediction uncertainty in an explicit and auditable manner.

Overall, the structural expert integrates residue-level sequence semantics, confidence-aware geometry, full-atom chemical-site interactions, peptide-level physicochemical properties, and availability-masked conformer statistics. Rather than treating all predicted structural evidence

as equally reliable, GHXTox continuously modulates its contribution using residue-level and peptide-level confidence signals. The following section describes the training, out-of-fold model-selection, calibration, and parameter-freezing protocols used to optimize the complete framework.

2.5 Training and Model-Selection Protocol

GHXTox consists of a task-adaptive 1D sequence expert and a sequence-enhanced 3D structural expert. Because the two experts differ substantially in input representation, parameter scale, and optimization behavior, they are not jointly optimized through a single end-to-end objective. Instead, expert-level representations are learned independently, after which probability calibration and top-level fusion are performed using training out-of-fold (OOF) predictions. This strategy allows each expert to be optimized under modality-appropriate settings while separating representation learning, checkpoint selection, probability calibration, fusion-parameter determination, and independent evaluation into clearly defined stages.

The training protocol follows three central principles. First, identical or highly similar peptides must not appear simultaneously in a training partition and the corresponding validation partition. Second, every prediction used for calibration, fusion-weight selection, or threshold determination must be generated by a model that was not trained on the corresponding sample. Third, Test1 and Test2 are reserved exclusively for final evaluation of the frozen model and are not used for checkpoint selection, calibrator fitting, fusion search, or decision-threshold optimization.

2.5.1 Group-Aware Five-Fold OOF Generation

The training set contains $N = 6387$ peptides. To reduce performance inflation caused by highly similar sequences, a fixed five-fold manifest is constructed by clustering the training sequences into homology groups with CD-HIT at 80% sequence identity. The fold-level homology groups were constructed through a separate CD-HIT pass at 80% sequence identity, distinct from the 0.90 threshold used for dataset-level redundancy removal in Section 2.1. The first operation determines which samples remain in the benchmark dataset, whereas the second determines which retained samples must remain in the same cross-validation fold. Let

$$\mathcal{D} = \{(\mathcal{S}_i, y_i, g_i)\}_{i=1}^{N}, \qquad N = 6387, \tag{66}$$

denote the training data, where $\mathcal{S}_i$ is the peptide sequence, $y_i \in \{0, 1\}$ is its toxicity label, and g_i denotes its sequence-homology group. The data are partitioned into five disjoint validation folds:

$$\begin{aligned} \mathcal{D} &= \bigcup_{k=1}^{5} \mathcal{D}_k, \qquad\qquad \mathcal{D}_k \cap \mathcal{D}_l = \varnothing, \quad k \neq l, \\ g_i = g_j &\Longrightarrow f(i) = f(j), \end{aligned} \tag{67}$$

where $f(i)$ denotes the fold assigned to sample i. All peptides belonging to the same homology group are therefore placed in the same fold, preventing closely related sequences from crossing the training–validation boundary.

For fold k, the model is trained only on the complement $\mathcal{D}_{-k} = \mathcal{D} \setminus \mathcal{D}_k$ and evaluated on the unseen fold $\mathcal{D}_k$. The OOF probability of sample $i \in \mathcal{D}_k$ is defined as

$$\mathcal{D}_{-k} = \mathcal{D} \setminus \mathcal{D}_k, \qquad \widehat{p}_i^{\,\mathrm{OOF}} = F_{\boldsymbol{\theta}^{(-k)}}\left(\mathcal{S}_i\right), \quad i \in \mathcal{D}_k, \tag{68}$$

where $\boldsymbol{\theta}^{(-k)}$ denotes the parameters learned exclusively from $\mathcal{D}_{-k}$. Each training peptide consequently receives exactly one probability from a model that did not observe that peptide during optimization.

For the 1D expert, full seed-specific OOF predictions are generated for the training set, and the three probability vectors are aligned by sample index and averaged:

$$\overline{p}_{i,\mathrm{1D}}^{\mathrm{OOF}} = \frac{1}{3} \sum_{s \in \{42,123,2025\}} \widehat{p}_{i,\mathrm{1D}}^{(s,\mathrm{OOF})}. \tag{69}$$

The resulting OOF ensemble scores are used to fit the Platt calibrator described in Section 2.3.4.

The structural expert uses the same fixed group-aware folds. Each fold-specific checkpoint produces probabilities only for its held-out validation fold:

$$\widehat{p}_{i,\mathrm{SE3D}}^{\mathrm{OOF}} = G_{\boldsymbol{\phi}^{(-k)}} \left(\mathcal{S}_i, \mathcal{X}_i, \mathcal{A}_i, \mathbf{q}_i, \mathcal{C}_i \right), \qquad i \in \mathcal{D}_k, \tag{70}$$

where $\mathcal{X}_i$, $\mathcal{A}_i$, $\mathbf{q}_i$, and $\mathcal{C}_i$ denote the residue geometry, full-atom chemical information, pLDDT confidence, and available conformer-ensemble statistics, respectively; $\mathcal{C}_i$ may be unavailable and is handled by the conformer-availability mask defined in Section 2.4.1.

Before combining the expert-level OOF predictions, the implementation verifies immutable sample indices (`source_index`), peptide sequences, toxicity labels, and fold assignments. Duplicate sample indices or mismatches in sequence, label, or fold assignment cause the fusion procedure to terminate rather than silently aligning prediction files by row order. OOF predictions are treated as **training-only model-selection assets**. Although OOF predictions are generated by models that did not observe the corresponding validation samples, parameters selected from the complete OOF table remain development-stage choices and must be evaluated on independent test sets.

2.5.2 Loss Functions and Class-Imbalance Handling

The 1D and 3D experts are optimized using separate objectives. This prevents uneven structural coverage, conformer availability, and confidence variation from directly perturbing the language-model optimization, and it avoids the large ProtT5 parameter space from dominating the smaller structure-specific residual modules.

Loss for the 1D expert. As described in Section 2.3.3, the sequence expert is trained using positive-class-weighted binary cross-entropy:

$$\mathcal{L}_{\mathrm{1D}} = -\frac{1}{B} \sum_{i=1}^{B} \left[w_+ y_i \log \sigma(z_i) + (1 - y_i) \log \left(1 - \sigma(z_i)\right) \right], \qquad w_+ = \frac{N_-}{N_+}, \tag{71}$$

where the class counts N_+ and N_- are independently calculated from each fold's training partition. The current 1D training pipeline does not use Borderline-SMOTE or sequence point-mutation augmentation. The weighted loss increases the contribution of the minority toxic class without introducing synthetic near-duplicate peptides.

Base geometric and chemical-site stage. The structural expert is optimized in stages. The full-atom chemical-site branch is first initialized from a previously trained pLDDT-aware geometric model, while the remaining backbone is frozen. This stage uses positive-class-weighted BCE:

$$\mathcal{L}_{\mathrm{chem}} = -\frac{1}{B} \sum_{i=1}^{B} \left[w_+ y_i \log \sigma \left(z_i^{\mathrm{base}} \right) + (1 - y_i) \log \left(1 - \sigma \left(z_i^{\mathrm{base}} \right) \right) \right], \tag{72}$$

where w_+ is defined as in Section 2.3.3 and is recalculated from each fold's training partition.

During training, pLDDT-dependent Gaussian coordinate perturbations are applied to reduce sensitivity to local errors in predicted structures. The perturbation scale for residue i is

$$\sigma_i = \sigma_0 \left[1 + \kappa \left(1 - q_i \right) \right], \qquad \sigma_0 = 0.12 \text{ Å}, \quad \kappa = 2.0, \tag{73}$$

and residue-wise perturbations are drawn as

$$\boldsymbol{\varepsilon}_i \sim \mathcal{N}\left(\mathbf{0}, \sigma_i^2 \mathbf{I}_3\right), \qquad \widetilde{\mathbf{x}}_i = \mathbf{x}_i + \boldsymbol{\varepsilon}_i. \tag{74}$$

The same residue-wise displacement is applied consistently to the corresponding backbone, functional-group, and chemical-site coordinates, thereby preserving their intra-residue relative geometry. No coordinate perturbation is applied during validation or inference. Low-pLDDT residues therefore receive stronger perturbations than high-confidence residues. This augmentation uses pLDDT to modulate the perturbation magnitude but does not treat pLDDT as an exact estimate of coordinate-error variance.

Conformer-ensemble residual stage. During conformer-residual training, the pLDDT-aware geometric backbone, internal sequence encoder, chemical-site branch, and base classifier are frozen. Only the residue-level conformer encoder, the global conformer encoder, the conformer-residual MLP, and the conformer logit-residual parameters are optimized. This stage is initialized from the trained chemical-site checkpoint. To avoid applying a second class-frequency correction on top of the existing base classifier, the current conformer-residual configuration uses unweighted BCE and disables coordinate perturbation, with a residual-magnitude penalty:

$$\mathcal{L}_{\text{conf}} = \text{BCEWithLogits}\left(z_{\text{SE3D}}, y\right) + \lambda_\Delta \frac{\sum_{i=1}^{B} c_i \left|\Delta z_{\text{conf},i}\right|}{\max\left(\sum_{i=1}^{B} c_i,\, 1\right)}, \qquad \lambda_\Delta = 0.005, \tag{75}$$

where the residual-magnitude term is computed only on samples with available conformer information. It limits large logit corrections and encourages the conformer branch to act as a conservative refinement rather than re-learning the complete toxicity decision boundary.

The final conformer-residual layer is zero-initialized, and its output is additionally controlled by the hard `conformer_available` mask. Zero initialization makes the initial model equivalent to the single-structure expert, whereas the hard mask guarantees a zero residual for samples without conformer evidence. Together, these mechanisms provide a stable incremental extension of the structural model.

2.5.3 Optimization, Checkpoint Selection, and Hyperparameter Settings

Both experts are optimized using AdamW [CITE: AdamW, Loshchilov and Hutter], but different learning rates, training durations, and checkpoint-selection criteria are employed. All random-number generators are initialized with fixed seeds. Data are shuffled only during training, whereas validation and inference loaders preserve deterministic ordering. Table 2 summarizes the optimization and model-selection settings of both experts.

Table 2: Main optimization and model-selection settings of the two GHXTox experts.

Module or stage	Batch size	Learning rate	Weight decay	Epochs	Checkpoint rule
1D LoRA-ProtT5 + Bi-Mamba	16	3×10^{-4}	1×10^{-2}	Fixed 15	No early stopping
3D chemical-site refinement	32	5×10^{-4}	1×10^{-4}	Max. 30	Best validation MCC; patience 8
Conformer residual refinement	32	1×10^{-3}	5×10^{-4}	Max. 50	Minimum validation loss; patience 8
Full-data structural refit	32	Stage-specific	Stage-specific	Fixed 8	Median fold-selected epoch

1D expert. The LoRA-ProtT5 and bidirectional Mamba expert is trained with a batch size of 16, a learning rate of 3×10^{-4}, a weight decay of 1×10^{-2}, a fixed duration of 15 epochs, dropout

of 0.1, FP16 mixed precision, and ProtT5 gradient checkpointing, using random seeds 42, 123, and 2025. This configuration yields approximately 13.1 million trainable parameters, comprising the LoRA updates and the downstream sequence-classification modules, whereas the pretrained base transformations and embeddings remain frozen. Every fold model and every full-data model is trained for a fixed 15 epochs. Validation-based early stopping and a separate learning-rate scheduler are not used. A fixed training budget ensures consistency across folds and random seeds and avoids differences in training duration caused by noisy fold-level validation metrics. After cross-validation, three models are independently trained on the complete training set using the same configuration. The deployed sequence probability is obtained by averaging these three full-data models rather than selecting the seed with the highest validation performance.

3D chemical-site stage. The full-atom chemical-site refinement uses a batch size of 32, a learning rate of 5×10^{-4}, a weight decay of 1×10^{-4}, a maximum duration of 30 epochs, early-stopping patience of 8 epochs, base structural dropout of 0.22, and chemical-site dropout of 0.1. This stage is initialized from a pLDDT-aware geometric checkpoint, and only the chemical-site branch is made trainable. Training is terminated when the validation MCC does not improve for eight consecutive epochs, and the checkpoint with the highest validation MCC is retained.

Conformer-residual stage. The conformer-ensemble residual stage uses a batch size of 32, a learning rate of 1×10^{-3}, a weight decay of 5×10^{-4}, a maximum duration of 50 epochs, early-stopping patience of 8 epochs, conformer-residual dropout of 0.25, a conformer hidden dimension of 32, an initial conformer weight of 0.25, and a residual L1 coefficient of 0.005. Validation loss is used instead of training accuracy to select the conformer checkpoint, reducing sensitivity to threshold fluctuations arising from limited conformer coverage. Only conformer-related modules are optimized, whereas the geometric and chemical backbone remains frozen.

During structural-expert training, gradients of all trainable parameters are clipped to a maximum norm of 5.0 ($\|\nabla_\phi \mathcal{L}\|_2 \leq 5.0$), limiting unusually large gradients caused by local residual corrections or coordinate perturbations.

When a full-data structural checkpoint is required, the number of refitting epochs is not selected using either independent test set. Instead, a fixed epoch count is obtained from the median of the fold-specific selected epochs. The recorded fold-level epochs are 8, 8, 14, 2, and 4, whose median is 8; the full-data structural refit is therefore trained for eight fixed epochs.

2.5.4 OOF-Based Calibration, Fusion-Weight Selection, and Threshold Freezing

All calibration, fusion-weight, and threshold decisions are made exclusively on training out-of-fold (OOF) predictions generated under the fixed group-aware five-fold partition; Test1 and Test2 are never used for any parameter selection.

1D calibration. The OOF probabilities of the three seeds are averaged per sample, and a two-parameter Platt calibrator is fitted on these OOF ensemble scores (Eq. (23)); the fitted parameters are frozen and applied at inference.

Fusion-weight and threshold. For a grid of 1D weights $w \in [0, 1]$ (step 0.02), the fused OOF score $p = w\, p_{\mathrm{1D}} + (1 - w)\, p_{\mathrm{SE3D}}$ is thresholded and the Matthews correlation coefficient (MCC) on training OOF is computed; the weight–threshold pair maximizing OOF MCC is frozen as the deployed operating point ($w = 0.72$, threshold 0.4702).

Protocol note. In the original selection the 3D side of the OOF fusion used the previous-generation fused OOF (0.62× old ProtT5-TextCNN + 0.38× 3D expert). Re-selecting on the current 3D-expert OOF yields $w = 0.72$, threshold 0.4702, with held-out Test2 unchanged and Test1 BACC 0.9354 → 0.9323 (see `runs/lora_mamba_full/protocol_audit.json`). The deployed operating point is $w = 0.72$, threshold 0.4702.

2.6 Evaluation Metrics

2.6.1 Threshold-Dependent Classification Metrics

2.6.2 Ranking and Probability-Calibration Metrics

2.7 Robustness, Uncertainty, and Statistical Analysis

2.7.1 Multi-Seed Reproducibility Analysis

2.7.2 Class-Stratified Bootstrap Confidence Intervals

2.7.3 Paired Bootstrap and Randomization Significance Testing

2.7.4 Subgroup and Structure-Quality Analysis

2.7.5 Frozen External Generalization Evaluation

2.8 Interpretability Analysis Protocol

2.8.1 Residue Masking and Integrated-Gradient Attribution

2.8.2 Chemical-Site Deletion Analysis

2.8.3 Expert Agreement and Conflict Analysis

3 RESULTS AND DISCUSSION

3.1 Performance Comparison with Existing Models

Table 3: Placeholder: performance comparison with existing models (mirrors Table 2 of the reference paper; replace with your own).

3.2 Exploration of the Model Architecture

Figure 3: Placeholder: performance comparison of different pretrained protein language models (mirrors Figure 2 of the reference; replace with your own).

Table 4: Placeholder: comparison of projection dimensions, kernel sizes and attention heads (mirrors Table 3 of the reference paper; replace with your own).

3.3 Ablation Experiments

3.3.1 Analysis of the Effectiveness of Multimodal Features

3.3.2 Contribution Analysis of the Cross-Modal Fusion Mechanism

3.4 Evaluation of Model Robustness and Generalization

3.4.1 Robustness Analysis Under Embedding and Sequence-Level Perturbations

3.4.2 Analysis of the Model's Generalization Ability on the Imbalanced Test Sets

3.5 Visualization of Feature Learning Using UMAP

3.6 Interpretability Analysis

3.6.1 Feature Importance Analysis

3.6.2 Attention-Based Interpretability Analysis

3.6.3 Mutation-Scan Analysis

3.7 Limitations and Future Directions

4 CONCLUSIONS

Data and Software Availability

Author Contributions

Acknowledgements

Conflict of Interest Disclosure

Figure 4: Placeholder: results of the ablation study (mirrors Figure 3 of the reference paper; replace with your own).

Table 5: Placeholder: complexity and performance comparison (mirrors Table 4 of the reference paper; replace with your own).

Table 6: Placeholder: additional complexity introduced by the added modules (mirrors Table 5 of the reference paper; replace with your own).

Table 7: Placeholder: K-fold cross-validation results (mirrors Table 6 of the reference paper; replace with your own).

Table 8: Placeholder: robustness under FGM adversarial perturbation (mirrors Table 7 of the reference paper; replace with your own).

Table 9: Placeholder: robustness under conservative substitution perturbations (mirrors Table 8 of the reference paper; replace with your own).

Table 10: Placeholder: prediction-probability stability under perturbations (mirrors Table 9 of the reference paper; replace with your own).

Table 11: Placeholder: comparison on the imbalanced independent test set (mirrors Table 10 of the reference paper; replace with your own).

Table 12: Placeholder: comparison on the external imbalanced test set (mirrors Table 11 of the reference paper; replace with your own).

Figure 5: Placeholder: feature visualization of different model stages (mirrors Figure 4 of the reference paper; replace with your own).

Figure 6: Placeholder: SHAP feature-importance analysis (mirrors Figure 5 of the reference paper; replace with your own).

Figure 7: Placeholder: SHAP-based contribution analysis (mirrors Figure 6 of the reference paper; replace with your own).

Figure 8: Placeholder: attention-based interpretability analysis (mirrors Figure 7 of the reference paper; replace with your own).

Figure 9: Placeholder: mutation-scan analysis (mirrors Figure 8 of the reference paper; replace with your own).